



USN

--	--	--	--	--	--	--	--	--	--

NMAM INSTITUTE OF TECHNOLOGY, NITTE*Off-Campus Centre of Nitte (Deemed to be University)***Second Semester M.Tech. (CYBER SECURITY) (CBCS) Degree Examinations**

May - June 2024

22CBS221 – SECURITY AND RESILIENCE

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing Two full questions from Unit – I & Unit – II each and One full question from Unit – III.

Unit – I		Marks	BT*	CO*	PO*
1. a)	Illustrate the future trends in C2 systems by taking stock of the state-of-the-art and by applying a methodological mix of analytical processes.	10	L*2	1	1
b)	Classify the Cloud types according their feasibility in defence and security.	10	L2	1	1
2. a)	Model the Information flows of a future multi-domain C2 system with a diagram.	10	L2	2	2
b)	Explain the Standards of Data Centers Based on Cloud Technologies and Requirements for Modern Data Centers.	10	L2	2	2
3. a)	Outline the Implementation of Cloud Technologies in Defence and Security.	12	L2	1	1
b)	Explain the Benefits of Applying Cloud Technologies for Defence and Security.	8	L2	2	2
Unit – II					
4. a)	Outline the Situational Awareness Challenges.	10	L2	3	1
b)	Explain the Network architecture of the smart grid with diagram.	10	L2	4	3
5. a)	Discuss the various Cyber Attacks on smart grids. Also explain the Blockchain Application for Cyber Resiliency.	10	L2	3	1
b)	Explain the requirements for the cyber security in the smart grids.	10	L2	3	1
6. a)	How the Generalized Net (GN)—Model is used in Radio Communication Protocol Between Transmitter and Receiver to detect the possible cyber-attack over the UAV?	10	L2	4	3
b)	What is Self-Discrepancy Theory? Explain the use of self-Self-Discrepancy Theory in cyber resilience.	10	L2	4	3
Unit – III					
7. a)	Outline the Internal Threats to Critical Infrastructure.	10	L2	5	2
b)	Explain the Cyber Range (CR) Capabilities.	10	L2	5	2
8. a)	How to Protect from Internal Threats? Explain.	10	L2	5	2
b)	Explain CR Interactions and AR/VR Continuum.	10	L2	5	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
